

Data Science

Chapter 1: Exploratory Data Analysis

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Analysis

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What is Exploratory Data Analysis (EDA)?

Before modeling, understand your data.

Definition

- Analyze datasets and summarize their main characteristics
- Use visualizations and descriptive statistics

Why is EDA important?

- Detect patterns and trends
- Identify outliers and missing values
- Check model assumptions
- Generate insights and hypotheses



Typical EDA techniques

- Data visualization (histograms, boxplots, scatter plots)
- Summary statistics (mean, median, variance)
- Correlation analysis

Key Message: *EDA transforms raw data into understanding and better decisions.*

Samples and Features in Tabular Data

ID	Age	Height (cm)	Math Score
1	15	168	85
2	16	172	78
3	15	165	92

- Each **row** represents one **observation** from the **sample**.
- Each **column** represents a **feature** (measured **variable**).
- The **population** is the complete group from which the sample was drawn (e.g., all students at Hochschule Kaiserslautern).

In machine learning, tabular data is often represented as a matrix:

$$X = \begin{bmatrix} x_{11} & \dots & x_{1d} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nd} \end{bmatrix} \in \mathbb{R}^{n \times d}$$

- n : number of **samples (observations)**
- d : number of **features (variables)**
- x_{ij} : value of feature j for sample i

Feature Types

- **Qualitative (categorical):** Describe properties or categories, e.g.
 - Gender
 - Nationality
 - Legal form of a company
- **Quantitative:** Numerical values or measured quantities (including units), e.g.
 - Number of children
 - Annual income in EUR
 - Wind speed in m/s

Distinction:

- **Discrete:** Only specific, separated values are possible; obtained by **counting**, e.g.
 - Number of faces in a coin or die toss experiment
 - Number of people waiting in a checkout line
 - Number of 1 EUR coins in a wallet
- **Continuous:** Any real value within intervals is possible; obtained by **measurement**, e.g.
 - Body height in m
 - Deviation of a thread diameter from its target value in μm
 - Arrival time of a train at the station in hh:mm:ss

Note: Measurements inevitably involve rounding (measurement accuracy, resolution). However, the better the measuring instrument, the more precise the value (e.g., the number of meaningful decimal places). This applies to both analog and digital displays.

Measurement Levels

- **Nominal:** Only equality or difference can be identified; no ordering is possible; always qualitative, e.g.
 - Gender
 - Postal code
 - Nationality
- **Ordinal:** Categories are distinguishable and can be arranged in a natural or meaningful order, e.g.
 - School grade ("very good" to "insufficient")
 - Intelligence quotient (IQ)
 - Ranking position in the Bundesliga
- **Metric (cardinal):** Expresses quantitative differences (not only order, but also distances); given as numbers/quantities (with units), e.g.
Distinction:
 - Temperature in K or $^{\circ}\text{C}$
 - Blood pressure in mmHG
 - Price in EUR
 - Running times (start/finish time) in h

Population

- The complete set of all individuals, objects, or observations of interest.
- It represents the entire group about which we want to draw conclusions.
- *Example:* All students at Hochschule Kaiserslautern, population of a country, ...

Sample

- A subset of the population that is actually observed or measured.
- It is used to estimate or infer properties of the population.
- *Example:* 200 students randomly selected or considered in a specific course.

Example:

A car is selected as part of a drug-enforcement inspection sample. We speak of a **random sample** if, for example, every tenth car is checked and we assume that cars arrive in random order. A **systematic sample** would be checking all red vehicles. An **arbitrary sample** would occur if the officer selects vehicles without any criteria.

Example: Sample from Statistics Course (WT 17/18)

Observed feature values (Group a, Group b, Group a & b):

- Age **X** in years (quantitative, discrete, metric)
- Body height **Y** in cm (quantitative, continuous, metric)
- Vocational training **B** with values {yes, no} (qualitative, nominal)
- Gender **G** with values {m, w} (qualitative, nominal)

Merkmalsausprägung: Alter [Jahre] Min 19 Max 28
Körperlänge [cm] Min 155 Max 195
Berufsausbildung {j/n}
Geschlecht {m/w}

		Urliste 70 Elemente																																										
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41		
Element#	m	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41		
Alter	x	22	21	23	24	24	27	19	19	21	22	21	26	22	24	20	23	21	22	20	24	21	22	22	20	21	26	24	28	25	25	21	21	20	25	22	20	21	20	23	22	28		
Körperlänge	y	187	178	170	170	171	193	168	160	163	161	160	164	174	160	163	193	184	185	163	182	185	170	180	182	185	182	183	169	187	183	165	176	180	172	170	173	174	189	180	190			
Berufsausb.	b	n	n	n	n	n	j	n	n	n	n	n	j	n	n	n	n	n	n	n	n	n	n	n	n	n	n	j	n	j	j	j	n	n	n	j	n	n	n	j	n	j		
Geschlecht	g	m	m	m	m	m	m	w	w	w	w	w	m	w	w	w	m	m	m	w	m	m	m	m	m	m	m	m	w	m	m	m	m	w	w	w	w	w	w	m	m	m		
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		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41		
		26	21	23	22	20	24	20	24	23	22	26	20	25	22	25	22	26	25	22	23	23	20	26	21	23	23	20	26	21	25	25	22	23	23	23	23	23	23	23	23	23	23	23
		184	176	176	188	185	182	183	182	185	180	175	195	176	165	180	170	185	178	184	155	178	176	183	185	174	176	170	185	165	160													
		j	n	j	n	n	n	n	n	n	j	n	j	n	n	n	j	n	n	j	n	j	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	
		m	m	m	m	m	m	w	w	w	w	w	m	w	w	m	m	m	m	w	m	m	m	m	m	m	m	m	w	m	m	m	m	w	w	w	w	w	w	w	m	m	m	

Urlisten, nach Zügen getrennt:

Urliste Zug a																																												
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Alter	x	22	21	23	24	24	27	19	19	21	22	21	26	22	24	26	23	21	22	20	24	21	22	22	20	21	26	24	28	25	25	21	21	20	25	22	20	21	20	23	22	28		
Körperlänge	y	187	178	170	170	171	193	168	160	163	161	160	164	174	160	163	193	184	185	163	182	185	170	180	182	185	182	183	169	187	185	165	176	180	172	170	173	174	189	180	190			
Berufsausb.	b	n	n	n	n	n	j	n	n	n	n	n	j	n	n	n	n	n	n	n	n	n	n	n	n	n	n	j	n	j	j	j	n	n	n	j	n	n	n	j	n	j		
Geschlecht	g	m	m	m	m	m	m	w	w	w	w	w	w	m	w	w	m	m	m	w	m	m	m	m	m	m	m	m	m	m	w	m	m	m	m	w	w	w	w	w	w	m	m	n

Urliste Zug b																																											
Element#	m	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	
Alter	x	26	21	23	21	20	24	20	19	23	20	22	26	26	20	25	22	26	25	22	23	23	23	20	24	26	21	25	25	22	23												
Körperlänge	y	184	176	176	188	185	182	183	182	185	160	175	195	176	165	180	170	185	178	184	155	178	176	183	185	174	176	170	165	160													
Berufsausb.	b	j	n	j	n	n	n	n	n	n	n	j	n	j	n	n	n	j	n	n	j	n	j	n	n	n	n	j	n	n	n												
Geschlecht	g	m	m	m	m	m	m	m	w	w	w	m	w	w	w	m	m	m	m	m	w	m	m	m	m	w	w	w	w														

Frequency

Each feature value $\{x_1, \dots, x_i, \dots, x_k\}$ has the **absolute frequency**

$$n_i := \text{absFreq}(\mathbf{X} = x_i), \quad n_i > 0, \quad \sum_{i=1}^k n_i = n, \quad k \leq n$$

and the **relative frequency**

$$h_i := \text{relFreq}(\mathbf{X} = x_i) = \frac{n_i}{n}, \quad 0 < h_i \leq 1, \quad \sum_{i=1}^k h_i = 1, \quad k \leq n$$

Definition: Absolute and Relative Frequency

The tables

x_1	x_2	$\dots$	x_k
n_1	n_2	$\dots$	n_k

with $\sum_{i=1}^k n_i = n$

x_1	x_2	$\dots$	x_k
h_1	h_2	$\dots$	h_k

with $\sum_{i=1}^k h_i = 1$

are called the **absolute** and **relative frequency distributions**, respectively.

Example: Sample from Statistics Course (WT 17/18)

The frequency tables for the samples are:

Urliste gesamt (Züge a & b)				Urliste 1 (Zug a)				Urliste 2 (Zug b)			
Alter	Körperlänge	Berufsausbildung	Geschlecht	Alter	Körperlänge	Berufsausbildung	Geschlecht	Alter	Körperlänge	Berufsausbildung	Geschlecht
70, 1,00	70, 1,00	70, 1,00	70, 1,00	41, 1,00	41, 1,00	41, 1,00	41, 1,00	29, 1,00	29, 1,00	29, 1,00	29, 1,00
x_i n_i h_i	y_i n_i h_i	b_i n_i h_i	g_i n_i h_i	x_i n_i h_i	y_i n_i h_i	b_i n_i h_i	g_i n_i h_i	x_i n_i h_i	y_i n_i h_i	b_i n_i h_i	g_i n_i h_i
19 3 3/70	155 1 1/70	1 17/70	m 47/70	19 2 2/41	160 3 3/41	1 9/41	m 27/41	19 1 1/29	155 1 1/29	1 8/29	m 20/29
20 10 10/70	160 5 5/70	n 53/70	w 23/70	20 6 6/41	161 1 1/41	n 32/41	w 14/41	20 4 4/29	160 2 2/29	n 21/29	w 9/29
21 11 11/70	161 1 1/70			21 9 9/41	163 3 3/41			21 2 2/29	165 2 2/29		
22 14 14/70	163 3 3/70			22 8 8/41	164 1 1/41			22 6 6/29	170 2 2/29		
23 9 9/70	164 1 1/70			23 3 3/41	165 1 1/41			23 6 6/29	174 1 1/29		
24 6 6/70	165 3 3/70			24 5 5/41	168 1 1/41			24 1 1/29	175 1 1/29		
25 8 8/70	168 1 1/70			25 3 3/41	169 1 1/41			25 5 5/29	176 5 5/29		
26 6 6/70	169 1 1/70			26 2 2/41	170 5 5/41			26 4 4/29	178 2 2/29		
27 1 1/70	170 7 7/70			27 1 1/41	171 1 1/41				180 1 1/29		
28 2 2/70	171 1 1/70			28 2 2/41	172 1 1/41				182 2 2/29		
	172 1 1/70				173 1 1/41				183 2 2/29		
	173 1 1/70				174 2 2/41				184 2 2/29		
	174 3 3/70				176 1 1/41				185 4 4/29		
	175 1 1/70				178 1 1/41				188 1 1/29		
	176 6 6/70				180 3 3/41				195 1 1/29		
	178 3 3/70				182 3 3/41						
	180 4 4/70				183 2 2/41						
	182 5 5/70				184 1 1/41						
	183 4 4/70				185 3 3/41						
	184 3 3/70				187 2 2/41						
	185 7 7/70				189 1 1/41						
	187 2 2/70				190 1 1/41						
	188 1 1/70				193 2 2/41						
	189 1 1/70										
	190 1 1/70										
	193 2 2/70										
	195 1 1/70										

For Group a & b combined, for example:

$$\mathbf{X} \in \{x_1 = 19, x_2 = 20, \dots, x_{10} = 28\}$$

$$\mathbf{Y} \in \{y_1 = 155, y_2 = 160, \dots, y_{27} = 195\}$$

$$\mathbf{B} \in \{b_1 = j, b_2 = n\}$$

$$\mathbf{G} \in \{g_1 = m, g_2 = w\}$$

$$\text{absFreq}(\mathbf{X} = x_2 = 20) = n_2 = 10$$

$$\text{relFreq}(\mathbf{Y} = y_4 = 163) = h_4 = \frac{3}{70}$$

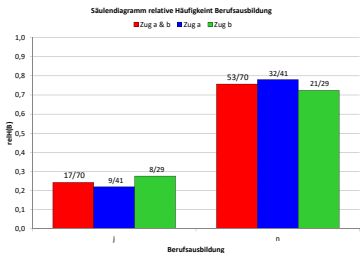
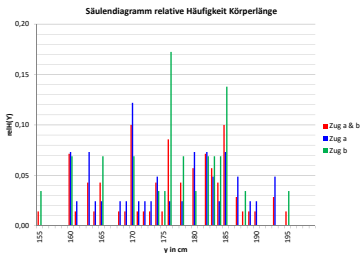
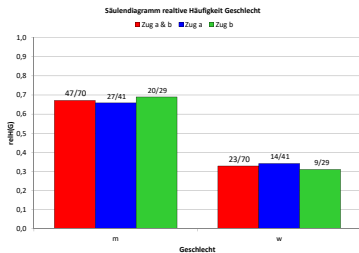
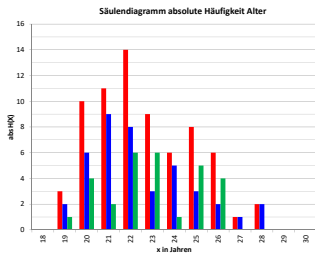
$$\text{relFreq}(\mathbf{B} = b_1 = j) = h_1 = \frac{17}{70}$$

$$\text{absFreq}(\mathbf{G} = g_2 = w) = n_2 = 23$$



Example: Sample from Statistics Course (WT 17/18)

Visualization of the features' frequencies as bar charts



Frequency & Distribution Function

For **metrically scaled feature values**, distances are defined, so that on a metric scale (typically on the horizontal axis / x -axis) the distances between feature values can be represented. On the vertical axis / y -axis, the corresponding frequencies can also be plotted metrically.

Definition: Frequency Function

The function

$$h(x) = \begin{cases} h_i & \text{if } x = x_i \\ 0 & \text{otherwise} \end{cases}$$

is called **frequency function** of the random variable $\mathbf{X}$.

Definition: Empirical Cumulative Distribution Function (ECDF)

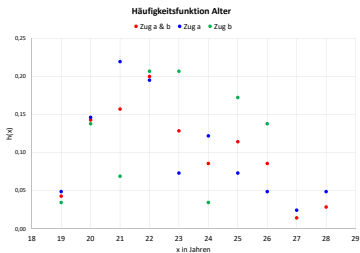
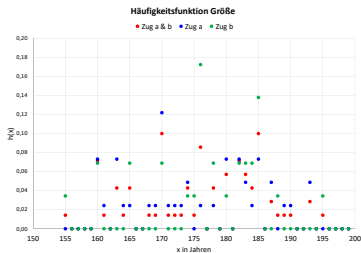
The function

$$H(x) = \sum_{x_i \leq x} h(x_i)$$

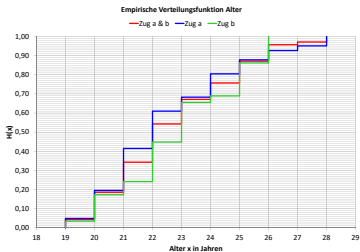
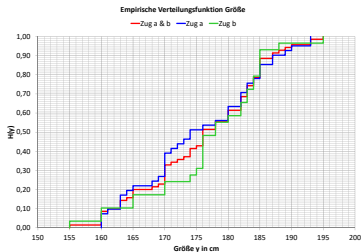
is called **empirical cumulative distribution function** of the random variable $\mathbf{X}$.

Example: Sample from Statistics Course (WT 17/18)

Frequency Functions



Empirical Cumulative Distribution Functions



Frequency & Distribution Function

The ECDF $H(x)$ has the following properties:

- 1 Everywhere at least right-continuous:

$$\lim_{\Delta x \rightarrow 0} H(x + \Delta x) = H(x), \quad \Delta x > 0$$

At the "jump points": Only right-continuous, i.e.

$$\lim_{\Delta x \rightarrow 0} H(x - \Delta x) \neq H(x), \quad \Delta x > 0$$

- 2 Monotonically increasing: $H(a) \leq H(b)$, if $a < b$, $a, b \in \mathbb{R}$

- 3 Limits:

$$\lim_{x \rightarrow -\infty} H(x) = 0, \quad \lim_{x \rightarrow +\infty} H(x) = 1$$

- 4 Difference: $H(b) - H(a) = \text{relFreq}(a < \mathbf{X} \leq b)$
- 5 Relationships between relative frequency and ECDF:

$$H(x) = \text{relFreq}(\mathbf{X} \leq x)$$

$$h(x) = H(x) - \lim_{\Delta x \rightarrow 0} H(x - \Delta x)$$

Earth Mover's Distance – Intuition

Intuitive idea:

- Two distributions = two piles of earth (or sand)
- Goal: reshape one pile into the other
- Minimize the required work

What defines work?

- How much mass is moved?
- How far is it moved?
- $\Rightarrow \text{Work} = \text{mass} \times \text{distance}$

Interpretation:

- Small EMD $\Rightarrow$ distributions are similar
- Large EMD $\Rightarrow$ distributions are very different

Also known as Wasserstein distance

Earth Mover's Distance – Formal Definition

Given:

- Two discrete distributions $P = (p_i)$ and $Q = (q_j)$
- Distance matrix $D = (d_{ij})$ with $d_{ij} = |p_i - q_j|$

Transport plan:

- γ_{ij} = mass moved from bin i to j
- Constraints:

$$\sum_j \gamma_{ij} = p_i$$

$$\sum_i \gamma_{ij} = q_j$$

$$\gamma_{ij} \geq 0$$

Definition:

$$EMD(P, Q) = \min_{\gamma} \sum_{i,j} \gamma_{ij} d_{ij}$$

Properties:

- Metric (under certain conditions)
- Sensitive to distribution structure
- Robust to small shifts

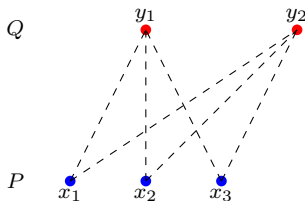
Earth Mover's Distance – Transport Plan Example

Distributions with 1D Histogram Bins

- $P = (0.3, 0.4, 0.3)$ with bins at positions $x = [1, 2, 3]$
- $Q = (0.5, 0.5)$ with bins at positions $y = [2, 4]$
- Distance: $d_{ij} = |x_i - y_j|$

Distance Matrix

$$D = \begin{pmatrix} 1 & 3 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}$$



Each entry d_{ij} corresponds to the distance between x_i and y_j .

Earth Mover's Distance – Transport Plan Example

Optimal Transport Plan γ

$$\gamma = \begin{pmatrix} 0.3 & 0 \\ 0.2 & 0.2 \\ 0 & 0.3 \end{pmatrix}$$

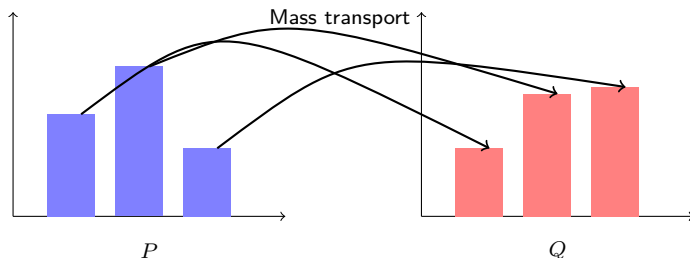
Interpretation:

- 0.3 from $x_1 \rightarrow y_1$
- 0.2 from $x_2 \rightarrow y_1$
- 0.2 from $x_2 \rightarrow y_2$
- 0.3 from $x_3 \rightarrow y_2$

Cost (EMD):

$$EMD = 0.3 \cdot 1 + 0.2 \cdot 0 + 0.2 \cdot 2 + 0.3 \cdot 1 = 1.0$$

Earth Mover's Distance – Visualization



Arrows indicate how probability mass is transported from P to Q .

Frequency Density & Histogram I

It is often useful to **condense** metric feature values into classes ("classified data"), or the data may be presented in this form from the start.

Class Intervals

The interval of the real axis covered by the possible feature values is subdivided into m different sections (intervals) by appropriately chosen class boundaries

$$\xi_0 < \xi_1 < \xi_2 < \dots < \xi_m$$

with not necessarily equal class widths

$$\Delta_i = \xi_i - \xi_{i-1} > 0, \quad i = 1, \dots, m$$

Relative Frequency of Class Intervals

$$h_i := \text{relFreq}(\xi_{i-1} < \mathbf{X} \leq \xi_i), \quad i = 1, \dots, m$$

Relative Frequency Distribution of Class Intervals

- Assigned to class upper boundaries (common):

ξ_1	ξ_2	$\dots$	ξ_m
h_1	h_2	$\dots$	h_m

with $\sum_{j=1}^m h_j = 1$

- Assigned to class midpoints (also possible):

$\frac{\xi_1 + \xi_0}{2}$	$\frac{\xi_2 + \xi_1}{2}$	$\dots$	$\frac{\xi_m + \xi_{m-1}}{2}$
h_1	h_2	$\dots$	h_m

with $\sum_{j=1}^m h_j = 1$

- From the relative frequency distribution of the class intervals, one obtains the distribution function of the class intervals $H_C(x)$
- If the function values of $H_C(x)$ are connected at the jump points by straight lines (polygons), an approximating distribution function $\overline{H}(x)$ is obtained as a polygon.
- In addition to the polygon as an approximation, there are sometimes other, usually "smooth" curves as "fits", which can be calculated and displayed by the software used.

Frequency Density of Class Intervals

If $H_C(x)$ is the distribution function of a feature grouped into class intervals with class boundaries $\xi_0 < \xi_1 < \xi_2 < \dots < \xi_m$ and $\overline{H}(x)$ is the approximating distribution function represented by a polygon, then the quotient

$$\frac{H_C(\xi_i) - H_C(\xi_{i-1})}{\xi_i - \xi_{i-1}} = \frac{h_i}{\Delta_i}, \quad i = 1, \dots, m$$

is called the (average) **frequency density** of the i -th class interval. The first derivative

$$\overline{h}(x) = \frac{d\overline{H}(x)}{dx}$$

is called the **frequency density function** and its graph the **histogram**.

Frequency Density & Histogram IV

Remarks

- The histogram is a piecewise constant function (similar to a bar chart whose bars are adjacent to each other) that is defined for all values.
- In contrast to the graph of the frequency function, in a histogram it is not the height of the bar, but rather its area

$$\left(\frac{h_i}{\Delta_i}\right) \cdot \Delta_i$$

that indicates the relative frequency of the class interval!

- The total area of the histogram is always

$$\sum_{j=1}^m \left(\frac{h_j}{\Delta_j}\right) \cdot \Delta_j = 1$$

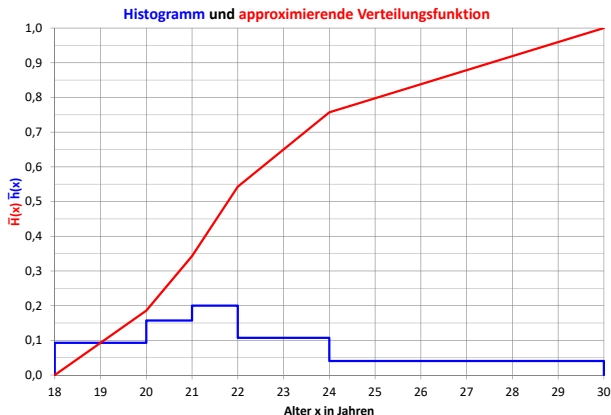
Example: Sample from Statistics Course (WT 17/18)

The feature value *age* can be condensed as follows, for example:

19	20	21	22	23	24	25	26	27	28
3	10	11	14	9	6	8	6	1	2



i	ξ_i	Δ_i	n_i	h_i	h_i/Δ_i	$H(\xi_i)$
0	18					0,000
1	20	2	13	13/70	39/420	0,186
2	21	1	11	11/70	66/420	0,343
3	22	1	14	14/70	84/420	0,543
4	24	2	15	15/70	45/420	0,757
5	30	6	17	17/70	17/420	1,000



Measures

A measure represents the most powerful form of condensing a statistical distribution into a single number.

Examples include:

- **Mean(s)**: Measure(s) for the "'central location'" of the feature values ("location measure")
- **Dispersion measure(s)**: Measure(s) for the variation of feature values around the location measure
- **Quantile(s)**: A quantile is the feature value in a sorted dataset at which a specified percentage of all values is less than or equal to it

Arithmetic Mean

The arithmetic mean can be calculated for metrically scaled feature values.

Definition: Arithmetic Mean

The quantity

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j \quad \text{Raw data list}$$

$$= \frac{1}{n} \sum_{j=1}^k n_j \cdot x_j \quad \text{Abs. frequency}$$

$$= \sum_{j=1}^k h_j \cdot x_j \quad \text{Rel. frequency}$$

is called the **arithmetic mean** or **mean value** of a statistical distribution.

Remark

Strictly speaking, the overline $\bar{}$ is a kind of "operator notation" in the sense that the result of averaging a quantity receives this overline. In this sense, $\overline{x^2}$ is the arithmetic mean of the squared feature values.

Example: Sample from Statistics Course (WT 17/18)

For Group a & b together it follows for the feature value *age* ($n = 70, k = 10$):

$$\begin{aligned}\bar{x}_{a \& b} &= \frac{1}{70} \sum_{j=1}^{70} x_j = \frac{1}{70} (22 + 21 + \dots + 22 + 23) = \frac{1.586}{70} \approx 22,657 \\ &= \frac{1}{70} \sum_{j=1}^{10} n_j \cdot x_j = \frac{1}{70} (3 \cdot 19 + 10 \cdot 20 + \dots + 2 \cdot 28) = \frac{1.586}{70} \approx 22,657 \\ &= \sum_{j=1}^{10} h_j \cdot x_j = \frac{3}{70} \cdot 19 + \frac{10}{70} \cdot 20 + \dots + \frac{2}{70} \cdot 28 = \frac{1.586}{70} \approx 22,657\end{aligned}$$

For both metric features *age* **X** and *body height* **Y** it follows:

Urliste gesamt (Zug a & b)			
Umfang	70 Elemente	Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste	22,657	Urliste	176,071
absH	22,657	absH	176,071
relH	22,657	relH	176,071

Urliste 1 (Zug a)			
Umfang	41 Elemente	Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste	22,488	Urliste	175,463
absH	22,488	absH	175,463
relH	22,488	relH	175,463

Urliste 2 (Zug b)			
Umfang	29 Elemente	Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste	22,897	Urliste	176,931
absH	22,897	absH	176,931
relH	22,897	relH	176,931

One can also compute the mean based on the means of the class intervals, e.g. ($k = 5$)

$$\bar{x} = \sum_{j=1}^5 h_j \cdot \bar{x}_j = \frac{13}{70} \cdot 19,769 + \dots + \frac{17}{70} \cdot 25,824 \approx 22,657$$

Note: Deviations may arise due to roundings.

i	ξ_i	Δ_i	n_i	h_i	$\bar{x}_i$
0	18				
1	20	2	13	13/70	19,769
2	21	1	11	11/70	21,000
3	22	1	14	14/70	22,000
4	24	2	15	15/70	23,400
5	30	6	17	17/70	25,824

Properties of the Arithmetic Mean

- ① Central property:

$$\sum_{j=1}^n (x_j - \bar{x}) = 0$$

- ② Shift by $a = \text{const.}$:

$$y_j = x_j + a \Rightarrow \bar{y} = \bar{x} + a$$

- ③ Scaling by $b = \text{const.}$:

$$z_j = b \cdot x_j \Rightarrow \bar{z} = b \cdot \bar{x}$$

- ④ Calculation from sub-series: The statistical series $\mathbf{X}$ is decomposed into m disjoint sub-series $\mathbf{X}_j$, $j = 1, \dots, m$, each with n_j elements and corresponding arithmetic means $\bar{x}_j$. Then we have

$$\bar{x} = \frac{1}{n} \sum_{j=1}^m n_j \cdot \bar{x}_j$$

e.g. $n = 70$, $m = 2$ (Group a & b), $n_1 = 41$, $n_2 = 29$:

$$\bar{x} = \frac{41}{70} \cdot 22,488 + \frac{29}{70} \cdot 22,897 \approx 22,656 \text{ (rounding inaccuracy)}$$

The median can be determined for metrically or ordinally scaled feature values.

Definition: Median

For a raw data list **sorted in ascending order**, the number x_{Med} with

$$\begin{aligned}x_{\text{Med}} &= x_{\frac{n+1}{2}} && n \text{ odd} \\&= \frac{1}{2} \left(x_{\frac{n}{2}} + x_{\frac{n}{2}+1} \right) && n \text{ even and } \mathbf{X} \text{ metrically scaled} \\x_{\frac{n}{2}} \leq x_{\text{Med}} \leq x_{\frac{n}{2}+1} &&& n \text{ even and } \mathbf{X} \text{ not metrically scaled}\end{aligned}$$

is called the **median** of the statistical series $\mathbf{X}$.

The median divides a raw data list sorted in ascending order in the middle, i.e. an equal number of observations lies to the left and right of the median, so it represents the “50% value”.

For the features *age* and *body height* (Group a & b, $n = 70$), the following holds (see next slide):

$$\begin{aligned}x_{\text{Med}} &= \frac{1}{2} \left(x_{\frac{70}{2}} + x_{\frac{70}{2}+1} \right) = \frac{1}{2} (x_{35} + x_{36}) = \frac{1}{2} (22 + 22) = 22 \\y_{\text{Med}} &= \frac{1}{2} \left(y_{\frac{70}{2}} + y_{\frac{70}{2}+1} \right) = \frac{1}{2} (y_{35} + y_{36}) = \frac{1}{2} (176 + 176) = 176\end{aligned}$$

S. Steidel & F. Wick

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[illegible]

Definition: Variance and Standard Deviation

The mean squared deviation from the arithmetic mean $\bar{x}$

$$s_X^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})^2$$

Raw data list

$$= \frac{1}{n} \sum_{j=1}^k n_j \cdot (x_j - \bar{x})^2$$

Absolute frequency

$$= \sum_{j=1}^k h_j \cdot (x_j - \bar{x})^2$$

Relative frequency

is called the **empirical variance**, or simply the **variance**, of the statistical series **X**. The positive root of s_X^2

$$s_X := +\sqrt{s_X^2}$$

is called the **empirical standard deviation**, or simply the **standard deviation**, of the statistical series **X**.

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Properties of Variance

- ① Non-negative:

$$s_X^2 \geq 0$$

- ② Shift by $a = \text{const.}$:

$$y_j = x_j + a \implies s_Y^2 = s_X^2$$

- ③ Scaling by $b = \text{const.}$:

$$z_j = b \cdot x_j \implies s_Z^2 = b^2 \cdot s_X^2 \text{ and } s_Z = |b| \cdot s_X$$

- ④ Simplified computation:

$$s_X^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})^2 = \frac{1}{n} \left(\sum_{j=1}^n x_j^2 \right) - \bar{x}^2 = \overline{x^2} - \bar{x}^2$$

- ⑤ Computation from sub-series: The statistical series $\mathbf{X}$ is decomposed into m disjoint sub-series $\mathbf{X}_j$, $j = 1, \dots, m$, each with n_j elements and corresponding arithmetic means $\bar{x}_j$ and variances s_j^2 . Then

$$s_X^2 = \underbrace{\frac{1}{n} \sum_{j=1}^m n_j \cdot s_j^2}_{\text{Inner variance}} + \underbrace{\frac{1}{n} \sum_{j=1}^m n_j (\bar{x}_j - \bar{x})^2}_{\text{Outer variance}}$$

Example: Sample from Statistics Course (WT 17/18)

Variance calculations

Urliste gesamt (Zug a & b)			
Umfang 70 Elemente		Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste 22,657		Urliste 176,071	
absH 22,657		absH 176,071	
relH 22,657		relH 176,071	
Varianz aus		Varianz aus	
Urliste 4,940		Urliste 89,838	
absH 4,940		absH 89,838	
relH 4,940		relH 89,838	

Urliste 1 (Zug a)			
Umfang 41 Elemente		Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste 22,488		Urliste 175,463	
absH 22,488		absH 175,463	
relH 22,488		relH 175,463	
Varianz aus		Varianz aus	
Urliste 5,372		Urliste 94,005	
absH 5,372		absH 94,005	
relH 5,372		relH 94,005	

Urliste 2 (Zug b)			
Umfang 29 Elemente		Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste 22,897		Urliste 176,931	
absH 22,897		absH 176,931	
relH 22,897		relH 176,931	
Varianz aus		Varianz aus	
Urliste 4,23		Urliste 82,685	
absH 4,23		absH 82,685	
relH 4,23		relH 82,685	

Combining outer and inner variance, $n = 70, m = 2$ (Group a & b),
 $n_1 = 41, n_2 = 29$, e.g. for *body height*:

$$\begin{aligned}
 s_Y^2 &= \underbrace{\frac{1}{70} (41 \cdot 94,005 + 29 \cdot 82,685)}_{\text{Inner variance, 89,315}} \\
 &+ \underbrace{\frac{1}{70} (41 \cdot (175,463 - 176,071)^2 + 29 \cdot (176,931 - 176,071)^2)}_{\text{Outer variance, 0,523}} \\
 &= 89,315 + 0,523 = 89,838
 \end{aligned}$$

Zusammenfassen Zug a & b					
Alter			Körperlänge		
Mittelwert		22,657	Mittelwert		176,071
Varianz innere		4,899	Varianz innere		89,315
äußere		0,041	äußere		0,523
Σ		4,940	Σ		89,838

Note on Variance: The Normalization Question

There is a **second definition of variance**, namely

$$s_X^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{x})^2$$

Raw data list

$$= \frac{1}{n-1} \sum_{j=1}^k n_j \cdot (x_j - \bar{x})^2$$

Absolute frequency

This definition normalizes by $n-1$ instead of n and provides, via s_X^2 , a "best possible" estimate for the true, unknown variance of the unknown population from which the sample was taken.

For inferential statistics, this second definition is therefore the "better" choice (unbiased point estimation).

However, both definitions can easily be converted into each other because they differ by the normalization ratio $\frac{n}{n-1}$.

Conclusion:

When reporting s_X^2 or s_X , it must be clear which normalization was used.

Further Dispersion Measures

There are additional dispersion measures that are not discussed further here:

- **Range:** Difference between the largest and smallest feature value:

$$\text{Range} := x_{\max} - x_{\min}$$

Applicable to metrically scaled feature values.

- **Mean absolute deviation:** Arithmetic mean of the absolute deviations of the feature values from the arithmetic mean $\bar{x}$

$$\text{MAD} := \frac{1}{n} \sum_{j=1}^n |x_j - \bar{x}|$$

Applicable to metrically scaled feature values.

- **Coefficient of variation:**

$$\text{VC}_X := \frac{s_X}{|\bar{x}|}, \quad \bar{x} \neq 0$$

Applicable to metrically scaled feature values.

Definition: Quantile

A number $x_{[q]}$ with $0 < q < 1$ is called the **q -quantile** if it partitions the statistical series $\mathbf{X}$ such that at least $100 \cdot q\%$ of its feature values are less than or equal to $x_{[q]}$, and at the same time at least $100 \cdot (1 - q)\%$ are greater than or equal to $x_{[q]}$, i.e.

$$\text{relFreq}(\mathbf{X} \leq x_{[q]}) \geq q \quad \text{and} \quad \text{relFreq}(\mathbf{X} \geq x_{[q]}) \geq 1 - q$$

- Typical, frequently used quantiles:

- Quantiles

- Lower quartile: $Q_1 = x_{[0,25]}$
 - Middle quartile: $Q_2 = x_{[0,50]} = x_{\text{Med}}$ (median)
 - Upper quartile: $Q_3 = x_{[0,75]}$

From the quartiles, the **five-number summary** is obtained:

- Dividing the data set into the four sections

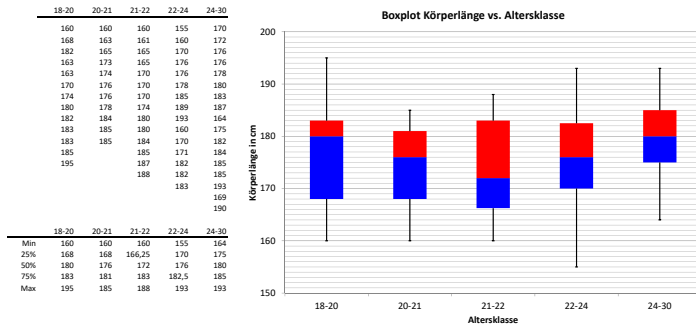
$$(x_{\min}, x_{[0,25]}, x_{\text{Med}}, x_{[0,75]}, x_{\max})$$

- Plotting them as a box plot (*Box-Plot*, *Box-Whisker-Plot*)

- Deciles: $x_{[0,1]}, x_{[0,2]}, \dots, x_{[0,9]}$
 - Percentiles: $x_{[0,01]}, x_{[0,02]}, \dots, x_{[0,99]}$

Example: Sample from Statistics Course (WT 17/18)

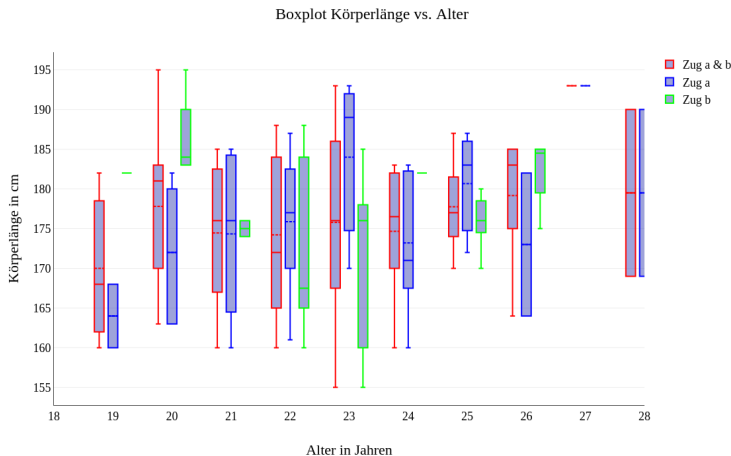
Box plots of *body height* vs. *age classes*



Note: The data are *not* sorted in ascending order.

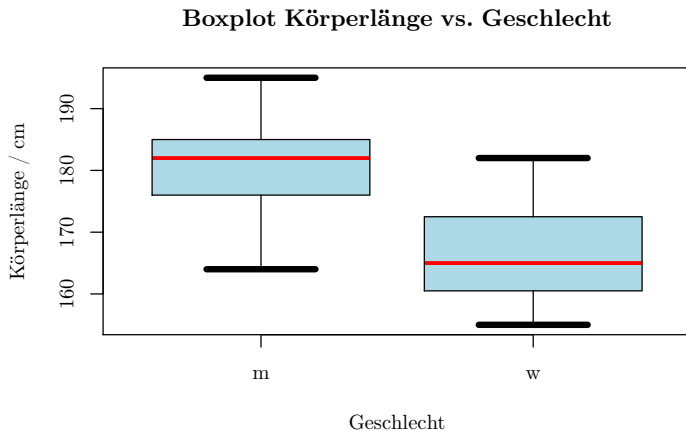
Example: Sample from Statistics Course (WT 17/18)

Box plots of *body height* vs. *age*, created with `http://plot.ly`



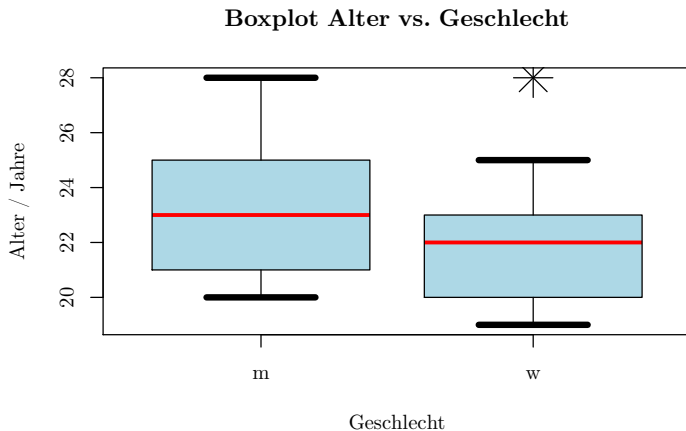
Example: Sample from Statistics Course (WT 17/18)

Box plots of *body height* vs. *gender*, created with R



Example: Sample from Statistics Course (WT 17/18)

Box plots of age vs. gender, created with R



Motivation: Multivariate Data

Up to now (*univariate*):

- Each feature considered **individually**
- Analysis of distributions, means, variances
- Relationships between features **ignored**

Problem:

- Important information is lost
- Dependencies and interactions remain hidden

Now (*multivariate*):

- Consider **multiple features simultaneously**
- Analyze **joint properties and relationships**
- Identify:
 - Correlations
 - Patterns and structures
 - Dependencies between features

Goal:

- Gain a **holistic understanding** of the data
- Enable better models and more informed decisions

General Principles

- Each row of a data matrix (from a population) can carry multiple features $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_m$, e.g.: all students of HSKL with feature $\mathbf{X}_1$: body height, feature $\mathbf{X}_2$: weight, feature $\mathbf{X}_3$: age, ...
- Considered below: two-dimensional statistics with the two features

$\mathbf{X}, \mathbf{Y}$.

The observed feature values are $(\mathbf{X} = x_i, \mathbf{Y} = y_i)$, written in short as (x_i, y_i) .

Definition: Scatter Plot

In a scatter plot, the observed feature values (x_i, y_i) are interpreted as points P_i and plotted in a y - x coordinate system.

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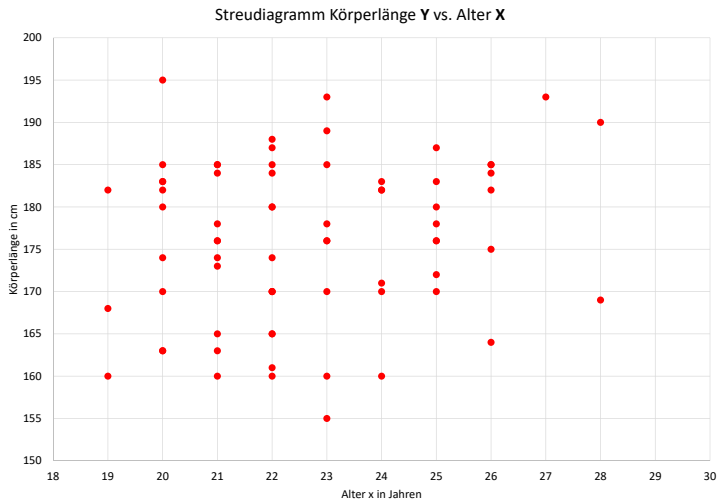
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Example: Sample from Statistics Course (WT 17/18)

For this sample, e.g. (Group a & b):



Contingency or Correlation Table I

Definition: Contingency or Correlation Table

The contingency or correlation table is the extension of one-dimensional frequency tables to two dimensions.

Absolute frequency with which the feature value combination (x_i, y_j) occurs according to the formed classes:

$$\text{absFreq}(\mathbf{X} = x_i \cap \mathbf{Y} = y_j) = n_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, l$$

$\text{absFreq}(x_i \cap y_j)$	y_1	y_2	$\dots$	y_j	$\dots$	y_l	Row sums
x_1	n_{11}	n_{12}	$\dots$	n_{1j}	$\dots$	n_{1l}	$n_{1\bullet}$
x_2	n_{21}	n_{22}	$\dots$	n_{2j}	$\dots$	n_{2l}	$n_{2\bullet}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_i	n_{i1}	n_{i2}	$\dots$	n_{ij}	$\dots$	n_{il}	$n_{i\bullet}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_k	n_{k1}	n_{k2}	$\dots$	n_{kj}	$\dots$	n_{kl}	$n_{k\bullet}$
Column sums	$n_{\bullet 1}$	$n_{\bullet 2}$	$\dots$	$n_{\bullet j}$	$\dots$	$n_{\bullet l}$	n

Contingency or Correlation Table II

- Absolute frequency with which x_i and y_j were observed, from row and column sums:

$$\text{absFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ arbitrary}) = n_{i\bullet} = \sum_{j=1}^l n_{ij}, \quad i = 1, \dots, k$$

respectively

$$\text{absFreq}(\mathbf{X} \text{ arbitrary} \cap \mathbf{Y} = y_j) = n_{\bullet j} = \sum_{i=1}^k n_{ij}, \quad j = 1, \dots, l$$

- Total number of observations:

$$n = \sum_{j=1}^l \sum_{i=1}^k n_{ij} = \sum_{i=1}^k n_{i\bullet} = \sum_{j=1}^l n_{\bullet j}$$

Contingency or Correlation Table III

- Of course, the contingency table can also be created for relative frequencies; then the following applies:

$$\text{relFreq}(\mathbf{X} = x_i \cap \mathbf{Y} = y_j) = h_{ij} = \frac{n_{ij}}{n}, \quad i = 1, 2, \dots, k, \quad j = 1, \dots, l$$

$$\text{relFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ arbitrary}) = h_{i\bullet} = \frac{n_{i\bullet}}{n} = \sum_{j=1}^l h_{ij}, \quad i = 1, \dots, k$$

$$\text{relFreq}(\mathbf{X} \text{ arbitrary} \cap \mathbf{Y} = y_j) = h_{\bullet j} = \frac{n_{\bullet j}}{n} = \sum_{i=1}^k h_{ij}, \quad j = 1, \dots, l$$

$$1 = \sum_{j=1}^l \sum_{i=1}^k h_{ij} = \sum_{i=1}^k h_{i\bullet} = \sum_{j=1}^l h_{\bullet j}$$

Example: Sample from Statistics Course (WT 17/18)

For this sample, e.g. (Group a & b):

absH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1					1											1										
	20				2					1				1				1	1	2								1	
	21		1		1		1						1	1		2	1				1	2							
	22		1	1			2			3				1				2			1	1	1	1					
	23	1	1							1						2	1					1			1		1		
	24		1							1	1								2	1									
	25									1		1					2	1	1		1		1						
	26					1									1					1									
	27																										1		
	28								1																	1			

relH($x_i \cap y_j$)		Y =																												
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195		
X =	19		1/70					1/70											1/70											
	20				2/70				1/70					1/70				1/70	2/70		1/70							1/70		
	21		1/70		1/70		1/70			1/70			1/70	1/70		2/70	1/70				1/70	2/70								
	22		1/70	1/70			2/70			3/70				1/70				2/70			1/70	1/70	1/70	1/70						
	23	1/70	1/70							1/70						2/70	1/70				1/70	1/70	1/70	1/70	1/70		1/70			
	24		1/70							1/70	1/70								2/70	1/70	1/70	1/70								
	25									1/70		1/70					2/70	1/70	1/70		1/70		1/70							
	26					1/70									1/70				1/70		1/70	2/70								
	27																										1/70			
	28								1/70																	1/70				

The $(k \times l)$ matrix is a (10×27) matrix, i.e. $i = 1, 2, \dots, 10$ and $j = 1, 2, \dots, 27$.

For the sake of clarity, zeros are not shown; for example, the entry $n_{53} = 0$ is therefore left blank.

Definition: Marginal Distributions

The one-dimensional distributions

$$\text{absFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ arbitrary}) = n_{i\bullet} = \sum_{j=1}^l n_{ij}, \quad i = 1, 2, \dots, k$$

$$\text{absFreq}(\mathbf{X} \text{ arbitrary} \cap \mathbf{Y} = y_j) = n_{\bullet j} = \sum_{i=1}^k n_{ij}, \quad j = 1, 2, \dots, l$$

$$\text{relFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ arbitrary}) = h_{i\bullet} = \frac{n_{i\bullet}}{n} = \sum_{j=1}^l h_{ij}, \quad i = 1, 2, \dots, k$$

$$\text{relFreq}(\mathbf{X} \text{ arbitrary} \cap \mathbf{Y} = y_j) = h_{\bullet j} = \frac{n_{\bullet j}}{n} = \sum_{i=1}^k h_{ij}, \quad j = 1, 2, \dots, l$$

are called the **marginal distributions** of the features $\mathbf{X}$ and $\mathbf{Y}$, respectively.

Remark

From the marginal distributions, the arithmetic means $\bar{x}$ and $\bar{y}$ as well as the variances s_X^2 and s_Y^2 can of course be computed.

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For this sample, e.g. (Group a & b):

absH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1					1											1										3
	20				2					1				1				1	1	2								1	10
	21		1		1		1						1	1		2	1				1	2						11	
	22		1	1			2			3				1				2			1	1	1	1				14	
	23	1	1							1						2	1								1			9	
	24		1							1	1								2	1								6	
	25									1		1				2	1	1		1			1					8	
	26					1				1			1		1		1	1		1	2		1					6	
	27																									1		1	
28								1																	1		1		
		1	5	1	3	1	3	1	1	7	1	1	1	3	1	6	3	4	5	4	3	7	2	1	1	1	2	1	70
		= n_{ij}																											

relH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1/70						1/70									1/70	1/70	2/70								1/70	3/70
	20				2/70					1/70					1/70			2/70	1/70	2/70		1/70	2/70					1/70	10/70
	21		1/70		1/70		1/70						1/70				2/70	1/70	1/70	2/70		1/70	2/70					1/70	11/70
	22		1/70	1/70			2/70			3/70				1/70			2/70				1/70	1/70	1/70	1/70				1/70	14/70
	23	1/70	1/70							1/70							2/70	1/70				1/70	1/70		1/70		1/70		9/70
	24		1/70							1/70	1/70						2/70	1/70	1/70		1/70					1/70		1/70	6/70
	25									1/70		1/70					2/70	1/70	1/70		1/70					1/70		1/70	8/70
	26					1/70						1/70			1/70		2/70	1/70	1/70		1/70	2/70		1/70				1/70	6/70
	27																										1/70	1/70	1/70
28								1/70																		1/70	1/70	1/70	2/70
		1/70	5/70	1/70	3/70	1/70	3/70	1/70	1/70	7/70	1/70	1/70	1/70	3/70	1/70	6/70	3/70	4/70	5/70	4/70	3/70	7/70	2/70	1/70	1/70	1/70	2/70	1/70	1,00
																						= h_1							

The marginal distributions are shown in red.

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Conditional Distributions

If one is interested in the distribution of one feature under the **condition** that the other has a specific fixed value, one obtains the so-called **conditional distribution**, e.g. the distribution of body height $\mathbf{Y}$ under the condition age $\mathbf{X} = 24$.

Notation: The presence of a condition is denoted by a vertical bar, " $|$ ", e.g. $\mathbf{Y}|\mathbf{X} = 24$.

Definition: Conditional Distribution

The $j = 1, 2, \dots, l$ one-dimensional distributions

$$\text{relFreq}(\mathbf{X} = x_i | \mathbf{Y} = y_j) = h_{i|y_j} = \frac{h_{ij}}{h_{\bullet j}} = \frac{n_{ij}}{n_{\bullet j}}, \quad i = 1, \dots, k$$

and the $i = 1, 2, \dots, k$ one-dimensional distributions

$$\text{relFreq}(\mathbf{Y} = y_j | \mathbf{X} = x_i) = h_{j|x_i} = \frac{h_{ij}}{h_{i\bullet}} = \frac{n_{ij}}{n_{i\bullet}}, \quad j = 1, \dots, l$$

are called **conditional distributions**.

Example: Sample from Statistics Course (WT 17/18)

For this sample, e.g. (Group a & b):

relH($y_j x_i$)		Y =																																		
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195								
X =	19		1/3					1/3		1/10				1/10				1/10	1/10	2/10		1/10						1/10							3/3	
	20				2/10					1/10				1/10				1/10	1/10	2/10		1/10						1/10							10/10	
	21		1/11		1/11		1/11						1/11	1/11		2/11	1/11				1/11	2/11												11/11		
	22		1/14	1/14			2/14			3/14				1/14				2/14				1/14	1/14	1/14	1/14									14/14		
	23	1/9	1/9							1/9						2/9	1/9					1/9				1/9		1/9						9/9		
	24		1/6							1/6	1/6								2/6	1/6														6/6		
	25									1/8		1/8					2/8	1/8	1/8		1/8			1/8										8/8		
	26					1/6										1/6				1/6		2/6												6/6		
	27																										1/1								1/1	
	28								1/2																	1/2									2/2	

relH($x_i y_j$)		Y =																																		
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195								
X =	19		1/5					1/1											1/4		2/3		1/5													
	20				1/2					1/5				1/3				1/3	1/4	2/3		1/5						1/1								
	21		1/5		1/2		1/2						1/1	1/3		1/3	1/3		1/3	1/4	2/3		1/5													
	22		1/5	1/1			1/2			2/5				1/3				2/3			1/3	1/5	1/2	1/1												
	23	1/1	1/5							1/5	1/1						1/3	1/3		2/3	1/3	1/5			1/1		1/2									
	24		1/5							1/5		1/1					1/3	1/3		2/4	1/3															
	25									1/5			1/1				1/3	1/3	1/3		1/3	1/5	1/2		1/1		1/1		1/2							
	26					1/1										1/1		1/3	1/3	1/3		1/4	1/3	1/5												
	27																					1/3	1/5													
	28								1/1																		1/1	1/2								
		1/1	5/5	1/1	2/2	1/1	2/2	1/1	1/1	5/5	1/1	1/1	1/1	3/3	1/1	3/3	3/3	3/3	4/4	3/3	3/3	5/5	2/2	1/1	1/1	1/1	2/2	1/1								
		Σ																																		

There are

- $k = 10$ conditional distributions $\text{relFreq}(\mathbf{Y} = y_j | \mathbf{X} = x_i)$ (top)
- $l = 27$ conditional distributions $\text{relFreq}(\mathbf{X} = x_i | \mathbf{Y} = y_j)$ (bottom)

Notably, choosing a condition for one variable can apparently restrict the possible values of the other variable; for example, under the condition $\mathbf{Y} = 155$, the feature $\mathbf{X}$ is not only restricted but even uniquely determined as $\mathbf{X} = 23$.

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Definition: Statistical Independence

If the joint distribution h_{ij} equals the product of the marginal distributions, i.e.

$$h_{ij} = h_{i\bullet} \cdot h_{\bullet j}, \quad i = 1, \dots, k, \quad j = 1, \dots, l$$

then $\mathbf{X}$ and $\mathbf{Y}$ are called **statistically independent**.

Remarks:

- 1 Under statistical independence, every entry of the contingency table h_{ij} equals the product of the row sum $h_{i\bullet}$ and the column sum $h_{\bullet j}$:

$$h_{i|y_j} = \frac{h_{ij}}{h_{\bullet j}} = \frac{h_{i\bullet} \cdot h_{\bullet j}}{h_{\bullet j}} = h_{i\bullet}, \quad i = 1, 2, \dots, k$$

$$h_{j|x_i} = \frac{h_{ij}}{h_{i\bullet}} = \frac{h_{i\bullet} \cdot h_{\bullet j}}{h_{i\bullet}} = h_{\bullet j}, \quad j = 1, 2, \dots, l$$

- 2 Intuitively, statistical independence means that the value of feature $\mathbf{X}$ does not allow any conclusion about the value of feature $\mathbf{Y}$ (and vice versa).
- 3 If $\mathbf{X}$ and $\mathbf{Y}$ are statistically dependent, this says nothing about a physical dependency (cause and effect).

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Definition: Covariance

The quantity

$$\begin{aligned} c_{XY} &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) \cdot (y_i - \bar{y}) \\ &= \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^l n_{ij} \cdot (x_i - \bar{x}) \cdot (y_j - \bar{y}) \\ &= \sum_{i=1}^k \sum_{j=1}^l h_{ij} \cdot (x_i - \bar{x}) \cdot (y_j - \bar{y}) \\ &= \bar{x} \cdot \bar{y} - \bar{x} \cdot \bar{y} \end{aligned}$$

is called the **empirical covariance**, or simply **covariance**, between the features **X** and **Y**.

Remark:

The covariance c_{XY} is zero if the features **X** and **Y** are (statistically) independent. Therefore, the covariance c_{XY} indicates statistical association between **X** and **Y**.

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Interpretation:

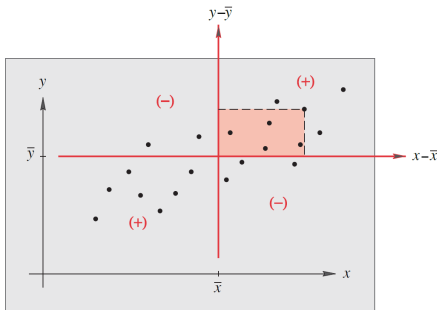


Figure: Illustration of covariance [Schira, Statistische Methoden der VWL und BWL, Pearson]

- Covariance is visualized using deviations from the mean, where each point forms a rectangle representing the product of deviations.
- Positive covariance arises when same-sign deviations dominate; opposite-sign deviations lead to negative covariance.
- Positive covariance means variables increase together, while negative covariance means one increases as the other decreases.

Principal Component Analysis (PCA) I

Step 1: Tabular Data

Height	Weight	Age
170	65	15
180	75	16
175	70	15

Step 2: Matrix Representation

$$X = \begin{bmatrix} 170 & 65 & 15 \\ 180 & 75 & 16 \\ 175 & 70 & 15 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$$

Step 3: Center the Data

Compute the mean of each feature: $\mu = [175 \quad 70 \quad 15.33] \in \mathbb{R}^{1 \times 3}$

Center the data by subtracting the mean:

$$X_c = X - \begin{bmatrix} \mu \\ \mu \\ \mu \end{bmatrix} = \begin{bmatrix} -5 & -5 & -0.33 \\ 5 & 5 & 0.67 \\ 0 & 0 & -0.33 \end{bmatrix}$$

$\implies$ Centering ensures that each feature has mean 0

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Principal Component Analysis (PCA) II

Step 4: Covariance Matrix

Covariance matrix of centered data:

$$\Sigma = \frac{1}{n} X_c^T X_c \approx \begin{bmatrix} 16.67 & 16.67 & 0.67 \\ 16.67 & 16.67 & 0.67 \\ 0.67 & 0.67 & 0.33 \end{bmatrix}$$

- Variance of Height = 16.67
- Variance of Weight = 16.67
- Variance of Age = 0.33
- Covariance Height-Weight = 16.67 (strong correlation)
- Age contributes very little

Summary:

$$s_{\text{Height}}^2 = 16.67, s_{\text{Weight}}^2 = 16.67, s_{\text{Age}}^2 = 0.33$$

$$c_{\text{Height,Weight}} = 16.67, c_{\text{Height,Age}} = 0.67, c_{\text{Weight,Age}} = 0.67$$

Principal Component Analysis (PCA) III

Step 5: Eigenvalues and Principal Components

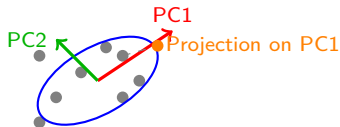
Solve the eigenvalue problem $\Sigma v_i = \lambda_i v_i$:

- $\lambda_1 \approx 33 \rightarrow \text{PC1}$ (largest variance)
- $\lambda_2 \approx 0.33 \rightarrow \text{PC2}$
- $\lambda_3 \approx 0 \rightarrow \text{PC3}$ (almost no variance)
- PC1 vector $\approx [0.71, 0.71, \varepsilon]$, ε small

Interpretation

Height and Weight dominate PC1 $\rightarrow$ almost all variance captured
Age has very little contribution $\rightarrow$ nearly PC3

Step 6: Project Data onto Principal Components

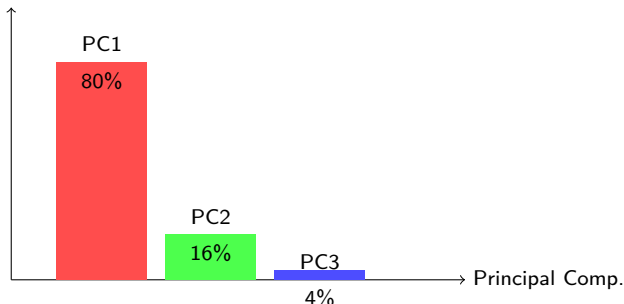


- PC1: direction of **maximum variance** (red)
- PC2: orthogonal to PC1 (green)
- Projecting data onto PCs reduces dimensionality while retaining most variance

Principal Component Analysis (PCA) IV

Explained Variance of Principal Components

Explained Variance (%)



- PC1, captures largest variation
- PC2, some variation, contributes moderately
- PC3, contributes very little, negligible
- Choosing first 1-2 PCs effectively reduces dimensionality
- This shows why PCA can **reduce dimensions without losing much information**

Definition: Correlation Coefficient

The quotient

$$r_{XY} = \frac{c_{XY}}{s_X \cdot s_Y}, \quad s_X > 0, s_Y > 0$$

is called the **empirical correlation coefficient** between the features **X** and **Y**.

Remarks:

- 1 r_{XY} can only be computed for metrically scaled features **X** and **Y**.
- 2 The quantity r_{XY} has the same sign as c_{XY} , so that it holds

$$-1 \leq r_{XY} \leq +1$$

with the interpretation

$$-1 \leq r_{XY} < 0 \quad \Leftrightarrow \quad \text{negative linear correlation}$$

$$r_{XY} = 0 \quad \Leftrightarrow \quad \text{no linear correlation}$$

$$0 < r_{XY} \leq +1 \quad \Leftrightarrow \quad \text{positive linear correlation}$$

- 3 Changes of scale (more precisely: linear transformations) do not affect r_{XY} .

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Correlation Coefficient II

- ④ Permutation of $\mathbf{X}$ and $\mathbf{Y}$ does not change $r_{\mathbf{X}\mathbf{Y}}$, i.e.

$$r_{\mathbf{X}\mathbf{Y}} = r_{\mathbf{Y}\mathbf{X}}$$

- ⑤ The correlation coefficient of the feature $\mathbf{X}$ with itself yields:

$$r_{\mathbf{X}\mathbf{X}} = \frac{c_{\mathbf{X}\mathbf{X}}}{s_{\mathbf{X}} \cdot s_{\mathbf{X}}} = \frac{s_{\mathbf{X}}^2}{s_{\mathbf{X}}^2} = 1$$

- ⑥ $r_{\mathbf{X}\mathbf{Y}}$ only indicates a *statistical tendency*. A *causal relationship* (in the sense of cause and effect) between the features $\mathbf{X}$ and $\mathbf{Y}$ does not necessarily exist; it may also be purely coincidental.

Example (Störche = storks & Geburten = births):

Land	Anzahl Störche je 100 km ²	Anzahl Geburten je 1000 Einwohner
	X	Y
Deutschland	3,5	0,8
Frankreich	3,9	0,9
USA	4,2	1,1
Brasilien	7,1	1,4
Argentinien	6,9	1,3
Rußland	4,5	1,2
Ägypten	12,8	1,8
Namibia	10,5	1,5
China	13,2	1,6
Indonesien	10,2	1,4
Indien	12,5	1,9

$$r_{\mathbf{X}\mathbf{Y}} = 0,92$$

- ⑦ $r_{\mathbf{X}\mathbf{Y}}$ is often referred to as the **Bravais-Pearson correlation coefficient**.

Example: Sample from Statistics Course (WT 17/18)

For this sample, for example (Group a & b), the following working table $x_i \cdot y_j \cdot h_{ij}$ is used to compute $\overline{x \cdot y}$:

$x_i \cdot y_j \cdot h_{ij}$		Y =													
		155	160	161	163	164	165	168	169	170	171	172	173	174	175
X =	19		3040/70					3192/70		3400/70				3480/70	
	20				6520/70									3654/70	
	21		3360/70		3423/70		3465/70						3633/70		
	22		3520/70	3542/70			7260/70			11220/70				3828/70	
	23	3565/70	3680/70							3910/70					
	24		3840/70							4080/70	4104/70				
	25									4250/70		4300/70			
	26					4264/70									4550/70
	27														
	28								4732/70						

$x_i \cdot y_j \cdot h_{ij}$		Y =													
		176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19				3458/70										
	20			3600/70	3640/70	7320/70		3700/70						3900/70	
	21	7392/70	3738/70				3864/70	7770/70							
	22			7920/70			4048/70	4070/70	4114/70	4136/70					
	23	8096/70	4094/70				4255/70				4347/70		4439/70		
	24				8736/70	4392/70									
	25	8800/70	4450/70	4500/70		4575/70			4675/70						
	26				4732/70		4784/70	9620/70							
	27											5320/70	5211/70		
	28														

Summing all entries yields $\overline{x \cdot y} = \frac{279.508}{70}$. With $\overline{x} = \frac{1.586}{70}$ and $\overline{y} = \frac{12.325}{70}$, it follows that $c_{XY} = \overline{x \cdot y} - \overline{x} \cdot \overline{y} = \frac{18.110}{4.900}$. With $\sigma_X^2 = \frac{24.204}{4.900}$ and $\sigma_Y^2 = \frac{440.205}{4.900}$, the correlation coefficient is

$$r_{XY} = \frac{c_{XY}}{s_X \cdot s_Y} = \frac{18.110/4.900}{\sqrt{24.204/4.900} \sqrt{440.205/4.900}} = \frac{18.110}{\sqrt{24.204 \cdot 440.205}} \approx 0,175$$

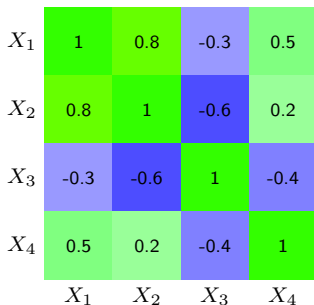
which indicates a weak positive correlation (a weak positive linear relationship).

Correlation Matrix and Heatmap

Correlation Matrix:

- A **correlation matrix** shows pairwise linear relationships.
- Values range from -1 (negative) to 1 (positive).
- Symmetric with ones on the diagonal.

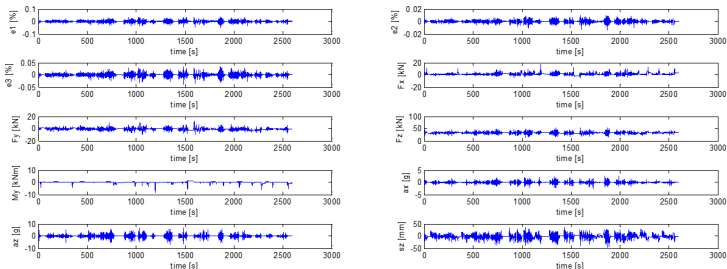
Heatmap Visualization:



Example: Load Data Analysis

Data example: Measurements in the area of the front wheel on a truck during a city trip

3 strains, 3 wheel forces, drive torque, 2 accelerations, spring displacement



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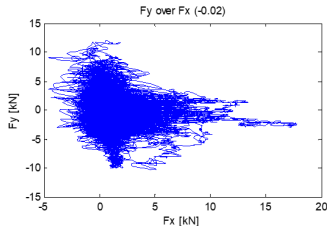
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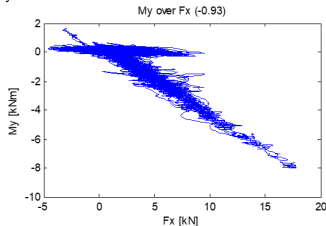
Example: Load Data Analysis

2D phase plots – a simple display option for visual evaluation

- **Fy over Fx** (very low negative correlation)
Lateral and longitudinal forces occur largely independently of each other.



- **My via Fx** (very high negative correlation)
The driving or braking torque leads to longitudinal forces. The longitudinal forces at **My = 0** are generated by unevenness.



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Example: Load Data Analysis

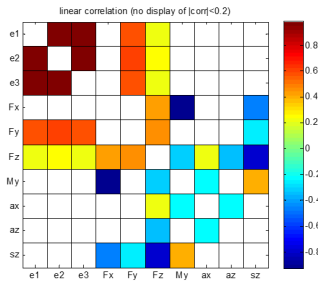
Linear (empirical) correlation coefficient – simple scalar variable for quantitative statements

Correlation between the variables $x_j(t), x_k(t)$:

$$r_{j,k} = \frac{\sum_i (x_j(t_i) - \bar{x}_j) \cdot (x_k(t_i) - \bar{x}_k)}{\sqrt{\sum_i (x_j(t_i) - \bar{x}_j)^2 \cdot \sum_i (x_k(t_i) - \bar{x}_k)^2}}, \bar{x}_j = \frac{1}{N} \cdot \sum_i x_j(t_i)$$

Application to the sample data:

- Strains very highly correlated with each other
- Strains with Fy highly correlated with Fz
- Fz and sz strongly negatively correlated
- Fz slightly correlated with Fx and Fy
- My strongly negatively / positively correlated with Fx / sz



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